

Introduction

Atrial Fibrillation (AFib) is a **dangerous arrhythmia** associated with **reduced or blocked conduction** in the heart's atria, leading to irregular heartbeats alongside **stroke and heart failure** (Dzeshka et al., 2015).

- Electrical remodelling in the heart perpetuates AFib and is thus the root issue.
- Increased reaction oxygen species (ROS)** lead to cellular damage and electrical remodeling (Pfenninger et. al, 2024).

Current treatments involve topical or oral medications and highly invasive surgery.

Hydrogels have seen success as therapies for various conditions

- They have been used to control ROS in the past (Cheng et al., 2024).
- An **electroactive, magnetically controlled hydrogel with ROS scavenging capabilities** could address the issues of AFib.
- Integration of magnetic nanoparticles using Fe²⁺ and Fe³⁺ provides for controllability of hydrogels. The influence of the adjustment of the ratio on properties of the hydrogel's electroactivity and scavenging ability is unknown.

Purpose

The purpose is to create an **optimal hydrogel** that can best target AFib by facilitating **electrical conduction** and **reducing ROS**.

Hypothesis + Variables

The optimal hydrogel should be prepared with a **1:2 ratio of Fe²⁺/Fe³⁺ ions** because the MNPs in the hydrogel are formed by the equation below:



The **stoichiometry** of this equation suggests that 1:2 is the best ratio as it **increases the number of MNPs** produced (Ganguly & Marvel, 2021). This will make it **more magnetically controllable** and perhaps **more electroactive**. The effects this will have on ROS control are unknown.

Independent variable: ratio of Fe²⁺ to Fe³⁺ ions

Dependent variable: magnetism, conductivity, ROS control
Control: N/A

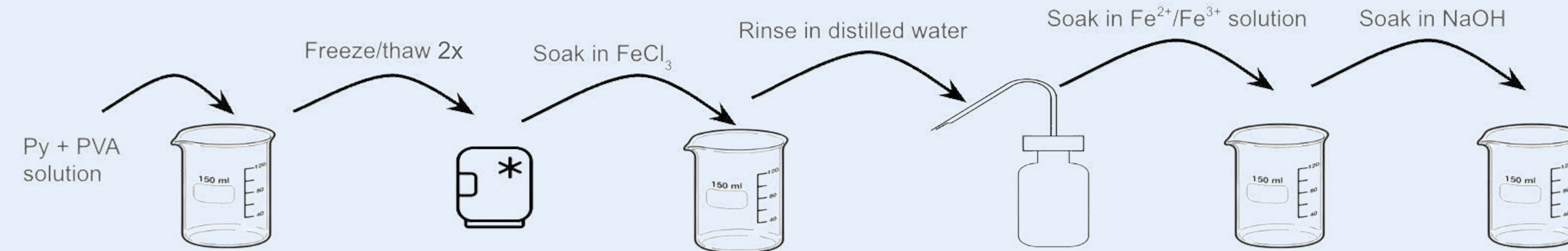
Determining the Optimal Ratio of Fe²⁺:Fe³⁺ for Magnetically Responsive and Electroactive Hydrogel With Reactive Oxygen Species Scavenging Capabilities

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Methods

*All photos and graphics taken/made by student researchers

Hydrogel synthesis:



Hydrogel synthesis is done via the process shown above. Pyrrole and polyvinyl alcohol were crosslinked through the freeze/thaw cycle, then embedded with magnetic nanoparticles which were deposited by the Fe²⁺/Fe³⁺ solution.

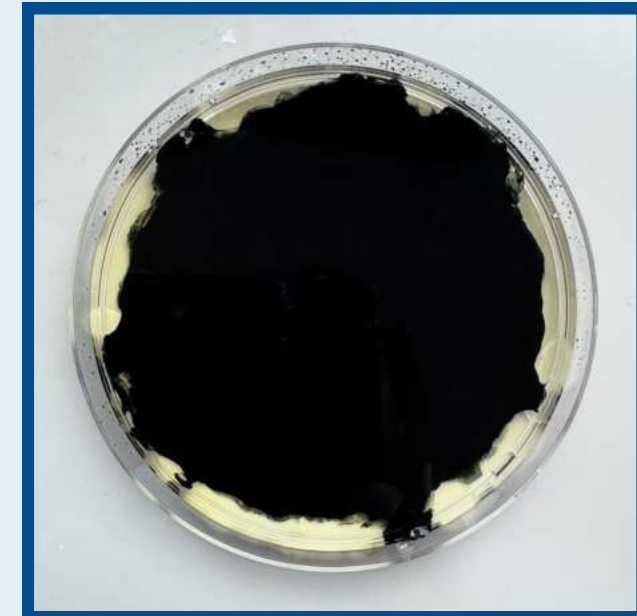


Fig. 1 - Hydrogel post-FeCl₃ bath



Fig. 2 - Hydrogel Sectioning

Hydrogel testing:

Magnetism

- Test the hydrogel for reaction to neodymium magnet.
- If it is attracted, it is a pass. If it is not, it is a fail.

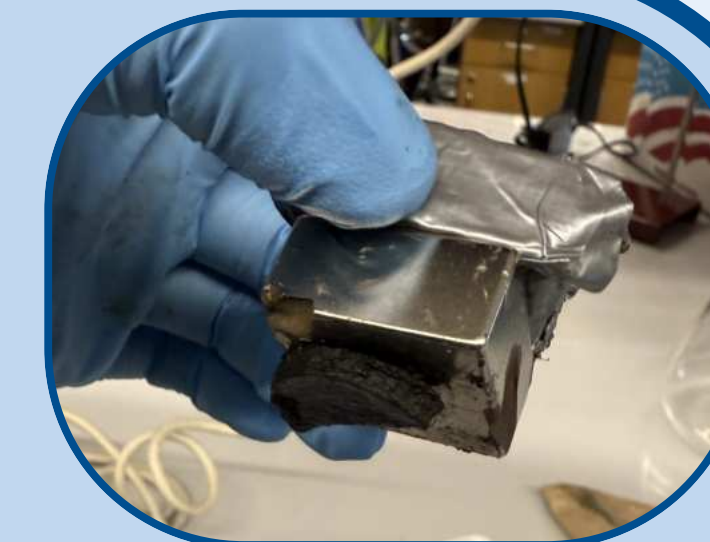


Fig. 3 - Magnetism Set-Up

Electroactivity

- Use multimeter to measure resistance of hydrogels at fixed distance apart.
- Measure resistance range from 10s to 60s.

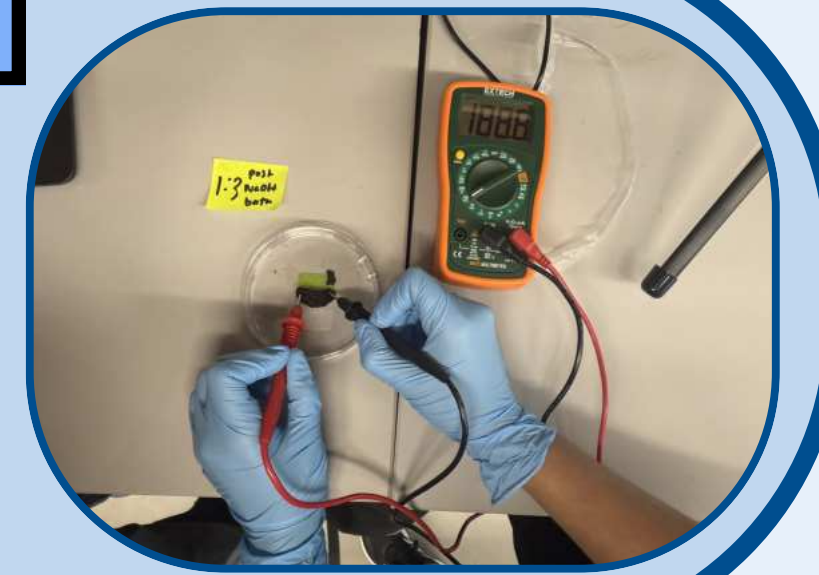


Fig. 4 - Electroactivity Set-Up

ROS Scavenging

- Submerge hydrogel in H₂O₂, then KMnO₄.
- H₂O₂ released by hydrogel into KMnO₄ will react to turn solution lighter.
- Measure absorbance with spectrophotometer - inversely proportional to H₂O₂ intake.



Fig. 5 - ROS Control Set-Up

Data and Results

Hydrogel	Resistance (Ω)								
	Trial 1			Trial 2			Trial 3		
	Highest	Lowest	Mean	Highest	Lowest	Mean	Highest	Lowest	Mean
1:1	1.443	1.340	1.3915	1.488	1.068	1.278	1.984	1.580	1.782
1:2	1.240	1.073	1.1565	1.488	1.200	1.344	1.166	0.786	0.976
1:3	0.946	0.659	0.8025	0.967	0.632	0.7995	1.175	0.883	1.029
1:5	1.997	1.010	1.5035	1.888	1.303	1.5955	1.995	1.635	1.815

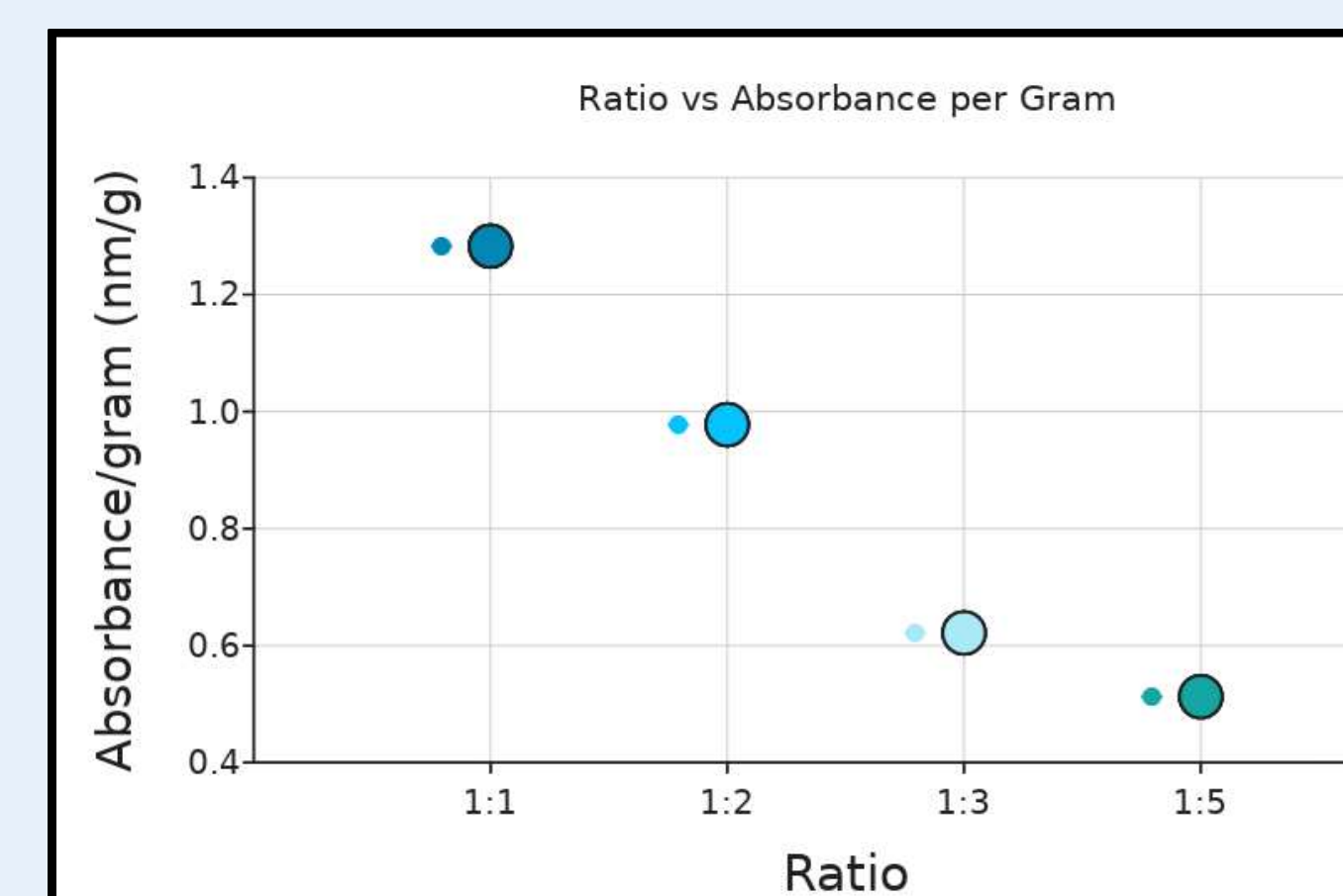
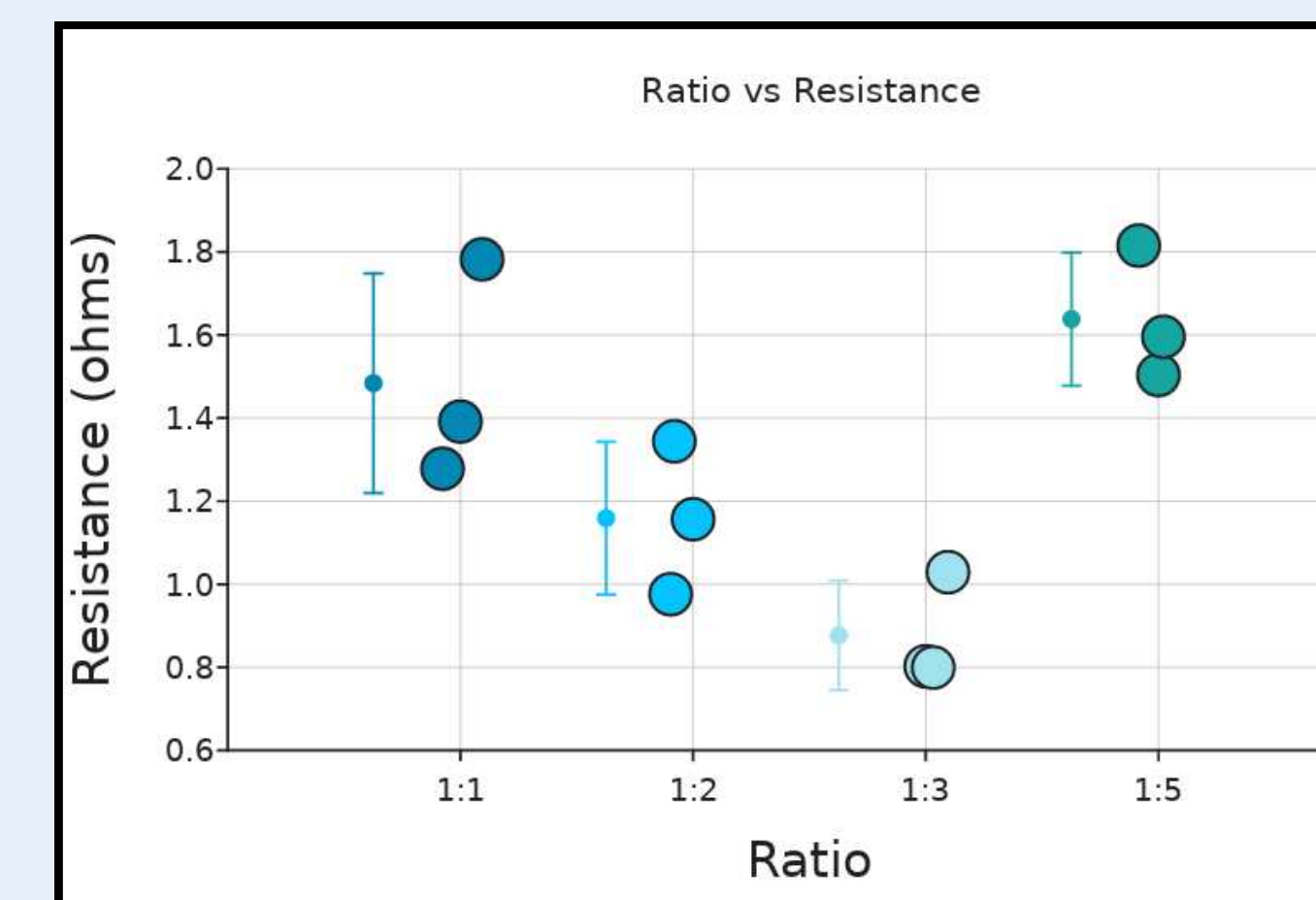
Table 2. Resistance of Hydrogels

Hydrogel	Absorbance	Mean Absorbance	Mass (g)
Baseline	1.289-1.296	1.2925	N/A
1:1	1.227-1.236	1.2315	0.78
1:2	1.214-1.230	1.2222	1.12
1:3	1.174-1.188	1.181	1.53
1:5	1.145-1.162	1.1535	1.52

Table 3. ROS Control of Hydrogels As Measured By Absorbance

Hydrogel	Pass/Fail
1:1	Pass
1:2	Pass
1:3	Pass
1:5	Pass

Table 1. Magnetism Test



Conclusion

Magnetism: As expected, all 4 hydrogels passed.

Absorbance/gram: As the amount of Fe³⁺ increases, the absorbance per gram **decreases**. In the following Kruskal-Wallis test, it was observed that the groups of 1:1 and 1:5 were statistically significant in difference (p=0.01), indicating that hydrogels with higher amounts of Fe³⁺ ions take up less ROS.

Electroactivity: The inherent trend in electroactivity is unclear, but the ANOVA test demonstrated that the iron ratio had an effect on electroactivity (p=0.01). The Tukey's test suggest strong differences between 1:1 and 1:3 (p=0.02) and 1:5 with 1:3 (p=0.01), as well as a less strong difference between 1:5 and 1:2 (p=0.06).

Tradeoff: Changing the iron ratio allows for **varying** hydrogel properties in which different ratios excel in different purposes.

- ROS Scavenging:** The 1:1 ratio appears most ideal, as it took up the most ROS.
- Electroactivity:** The 1:2 and 1:3 ratios were least resistant and therefore most conductively viable.

Implications & Future Work

Changing the ratio had significant alterations on the properties of hydrogels.

- Generally, the data suggests that varying iron ratios create inherent tradeoffs between conductivity and ROS control, indicating that the optimal hydrogel depends on the degree of usage.

Future studies should aim to:

- Test additional ratios for further optimization
- Use more precise conductivity and ROS measurement methods
- Analyze the structure of hydrogels with SEM or FTIR
- Quantitatively test for magnetic response

References

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